

Personalized Image Aesthetics Assessment

with Rich Attributes (PARA)

CVPR 2022 – Paper Summary

James Hsu – July 18th 2025

Motivation

- Traditional Image Aesthetic Assessment (IAA) assumes a universal notion of beauty
- Real-world aesthetics are subjective and personalized
- No existing dataset supports fine-grained personalized modeling
- Need for both **objective** and **subjective** annotations to enable personalized learning

PARA Dataset Overview

- **PARA** = *Personalized Aesthetics with Rich Attributes*
- 31,220 images, 438 unique subjects
- Each image rated by ~25 different individuals

Objective Attributes (Image-centric)

Each image is labeled with 9 visual attributes:

- Aesthetics
- Quality
- Composition
- Color
- Depth of Field
- Content
- Light
- object emphasis
- Scene categories

Subjective Attributes (User-centric)

Each user-image pair includes 4 subjective labels:

- Emotion
- Difficulty of judgement
- Content preference
- Willingness to Share

Users also provided anonymized background info:

- Age
- Gender
- Education level
- Art experience
- Photography experience
- Personality traits (Big-Five personality traits)

Key Statistical Insights

- Subjective attributes strongly correlate with aesthetic ratings
- Inter-rater disagreement is explainable via user characteristics
- Content preference, emotion, and sharing willingness are major factors for personalized taste

PARA Sample Image Example

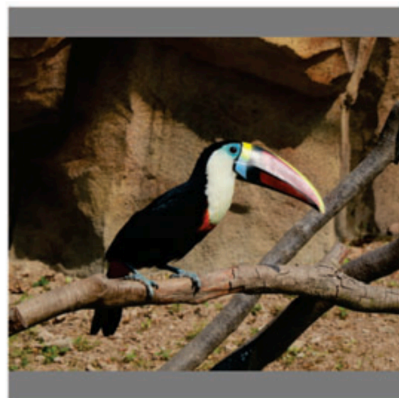


Image code: 175061040

3/80

(Click image to zoom in)

Tips:

1. Please carefully browse the image to be labeled on the left, and complete the annotation dimension on the right carefully. If the picture is not clear, you can click on the image to zoom in and check out the details. In addition, we have reference under each tag or on the right-up corner. Please check if any problem.

2. During the annotation process, please do not use "right-click" to return to the previous page, or force a refresh of the current page. Otherwise the submitted data will lose. If you need to return to the previous picture, please click the button "Return" to go back to the previous image and revise your choices.

3. There is no absolute correct answer in this task. Please finish all tags and double-check your answer and then click "submit" button to move on to the next image.

(Note: Accuracy of image quality dimension is 0.1, and image aesthetics score is 0.5. Please slide to label instead of click.)

1. Composition
Rule of third, symmetric, balanced element

☒ Much worse ☐ Worse ☐ Equal ☐ Better ☐ Much better

2. Light
Proper exposure, use of light

☐ Much worse ☐ Worse ☐ Equal ☐ Better ☐ Much better

3. Color
Colorfulness, color harmony, vivid color

☐ Much worse ☐ Worse ☐ Equal ☐ Better ☐ Much better

4. Depth of Field
Object emphasis or properly use the depth of field technique

Object emphasis: ☐ Yes ☐ No ☐ Much worse ☐ Worse ☐ Equal ☐ Better ☐ Much better

5. Content
Interestingness, appealing

☐ Much worse ☐ Worse ☐ Equal ☐ Better ☐ Much better

6. Image quality score
Noise, sharpness, exposures, contrast

Much worse

Worse

Equal

Better

Much better

3.0

7. Aesthetics score
Comprehensive aesthetic judgement

Much worse

Worse

Equal

Better

Much better

3.0

8. Emotion
Emotion feeling or transmission after browsing image

☐ Amusement ☐ Excitement ☐ Contentment ☐ Awe ☐ Disgust ☐ Sadness ☐ Fear ☐ Neutral

9. Difficulty
Difficulty of making aesthetics assessment judgement

☐ Easily ☐ Neutral ☐ Hard

10. I like the content of this photo

☐ Strongly disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly agree

11. Willingness of sharing this photo to social media

☐ Strongly disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly agree

Return

Submit(3s)

(Example image with annotation attributes)

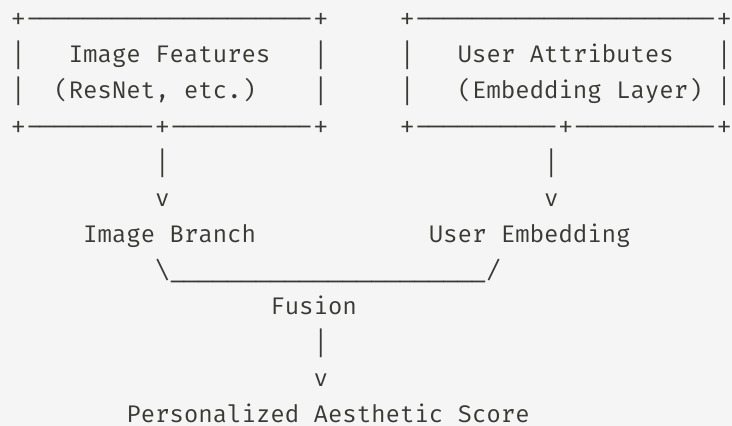
Conditional PIAA Model

A two-branch neural model:

1. **Image encoder** (e.g., ResNet) extracts image features
2. **User encoder** converts personal attributes into embeddings

These are fused to predict a **personalized aesthetic score**
(conditioned on user profile)

Model Architecture (Simplified)



Training Objective

The model is trained to minimize Mean Squared Error (MSE) between predicted and true scores.

Loss Function:

$$\mathcal{LMSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Where:

- y_i : Ground-truth rating (from user i)
- $\hat{y}_i$: Model prediction for user i
- N : Total number of user-image pairs

Experimental Results

- Models trained with user attributes consistently outperform user-agnostic baselines
- Adding even partial user info improves performance
- Incorporating subjective attributes like emotion boosts prediction accuracy

Dataset Comparison Table

Dataset	Personalized?	#Images	#Users	Attributes
AVA	 No	250k+	~500k	Aesthetic score only
AADB	 No	10k	N/A	11 aesthetic attributes
PARA	 Yes	31,220	438	Subjective + Objective

Contributions

- PARA: First large-scale dataset with rich subjective + objective labels
- In-depth analysis of how personal traits affect aesthetic judgment
- Introduced a conditional model that personalizes aesthetic prediction

Reference

Yang, Yuzhe, et al. "Personalized Image Aesthetics Assessment with Rich Attributes" CVPR 2022 Paper PDF